A

DATA SCIENCE FOR FINANCIAL MARKETS

REPORT
On

IND STOCKS CLONE

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CANDIDATE DECLARATION

We hereby certify that the work on the project entitled, ”**Project Name - IND Stocks clone**”, in partial fulfillment of requirements for the award of Degree of **Bachelor of Technology** in School of Engineering and Technology at BML Munjal University, is an authentic record of our own work carried out during a period from September 2025 November 2025 under the supervision of Dr. Hirdesh Kumar Pharasi .

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SUPERVISOR'S DECLARATION

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Faculty Supervisor Name: Dr.Hirdesh Kumar Pharasi

Signature:

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We would like to express our thanks profusely to thank Dr. Hirdesh Kumar Pharasi, for stimulating us from time to time. We would also like to thank the entire team at BML Munjal University. We would also thank my friends who devoted their valuable time and helped me in all possible ways toward successful completion.

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ABSTRACT

This project will develop an enhanced stock market analytics platform, inspired by the website IND Stocks, but based on much deeper and richer data-driven insights that link stock price movements with market sentiment to interpret investor behavior and market volatility. Using Yahoo Finance as the main data source, the system pulls in historical and real-time data for Indian equities to study return, correlation, and volatility patterns in the stock market.

The platform constructs correlation matrices in order to extract meaningful relationships between assets and applies RMT to filter out random noise, hence spotlighting genuine market structures like sectoral linkages and collective movements. Sentiment analysis on financial news and social media further refines these insights by incorporating investor mood into the modeling of market behavior.

Instead of allowing for live trading, the focus is more on explainability and understanding why markets behave in certain ways by connecting price dynamics and sentiment trends. Built using Python and Flask, the system offers an intuitive, research-oriented dashboard that helps users visualize volatility, correlation, and sentiment-driven market stress in a clear, interpretable way.

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CHAPTER 1 : INTRODUCTION

1.1 Background and Motivation

Financial markets have grown more complex in recent years, influenced by a multitude of factors, including price fluctuations, investor behaviour, volatility regimes and global events. Existing tools tend to analyze each category in isolation, making it difficult for beginners and non-experts to understand how they relate to each other.

Recently, tools based on advanced quantitative methods (GARCH volatility modeling, Random Matrix Theory (RMT), Deep-Learning Sentiment Analysis (FinBERT)) have become popular, but their technical difficulty and dispersed integration into observable actions makes them inaccessible to the average user.

This project builds on the need for a single, integrated system where users can poke around to examine market behaviour, understand advanced indicators and ultimately gain knowledge of how sentiment, correlations, volatility, and stress signals relate in practice.

1.2 Importance of the Topic

In an era of rapidly shifting markets, sentiment shifts influenced by social media and stress events, understanding market volatility and investor sentiment has become increasingly important.

This is important because:

- volatility spikes affect trading, risk management, and investment decisions.

- sentiment-driven markets require tools that capture psychological components in addition to price movements.

- correlation structure and market co-movement affect diversification, hedging, and portfolio risk.

- beginner user-friendly tools are scarce because most financial analytics platforms are too complex for an introductory learning experience.

By tackling these issues, this project will aid users from students to those aspiring to become analysts in understanding markets with a greater degree of completeness.

1.3 Scope of the Project

The project centers on developing a financial analytics dashboard that integrates:

- Live price data via Yahoo Finance
- GARCH(1,1) volatility forecasts
- FinBERT sentiment analysis extracted from news/social sources
- Correlations, correlation matrices and heatmaps
- Random Matrix Theory to filter noise and stress
- Sentiment-adjusted correlations
- Eigenvalue spectrum visualizations
- Automated insights in plain language

The scope of the project includes fetching data, backend analytics, building models, and a fully interactive frontend.
While the system can provide insights, it does not execute any trading or financial recommendations.

1.4 Structure of the Report

This project report is organized into the following chapters:

- **Chapter 1 – Introduction**
  Provides the background, motivation, importance, scope, and chapter outline.

- **Chapter 2 – Literature Review**
  Reviews volatility modelling, sentiment extraction, correlation analysis, RMT, and relevant research.

- **Chapter 3 – Problem Statement**
  Clearly defines the problem addressed by the project and the need for an integrated, beginner-friendly analytics system.

- **Chapter 4 – Methodology**
  Describes the step-by-step approach including data collection, sentiment scoring, GARCH volatility modelling, correlation computation, RMT application, and sentiment-adjusted correlation methodology.

- **Chapter 5 – System Design and Implementation**
  Explains the backend APIs, modules developed, frontend dashboard components, caching, and full architecture.

- **Chapter 6 – Results and Analysis**
  Presents all visualisations, model outputs, insights, and interpretation of market behaviour.

- **Chapter 7 – Conclusion and Future Work**
  Summarizes contributions, findings, limitations, and future enhancements.

CHAPTER 2: LITERATURE REVIEW

2.1 Areas Researched

Financial markets are tremendously complex systems, and many researchers have tried to understand how stocks move together, how volatility behaves, and how news or sentiment affects prices. This section summarizes the most important ideas from past studies that relate to the objectives of this project.

1. Understanding Correlations Between Stocks

Many early studies demonstrated that the correlation matrix of stock returns is full of noise. Laloux et al. [1] and Plerou et al. [2], for instance, applied the RMT approach to prove that only a few correlations are meaningful, while the majority of them come from randomness. Bouchaud and Potters [3] explained how RMT can be of assistance in identifying true market patterns such as the overall market movement or sector clusters.

The researchers also presented ways to "clean" or denoise the correlation matrix so that it becomes more reliable. In particular, Ledoit and Wolf [4] developed shrinkage estimators that combine sample data with a structured estimate to reduce error. Those methods have since entered common use to enhance risk management and portfolio decisions.

2. Volatility and Its Behaviour

Volatility clustering is one of the most widely accepted facts in financial time series. Engle [5] introduced the ARCH model, which was improved by Bollerslev [6] to the widely used GARCH model. These models show how periods of high volatility tend to follow one another.

This is important because volatility affects correlations and market stability, and many recent studies combine volatility models with correlation analysis.

3. Role of News and Sentiment

Another large body of research demonstrates that sentiment from news and social media influences market prices. Tetlock [7] determined negative news sentiment tends to be associated with temporarily declining markets. Antweiler and Frank [8] found that online stock forum messages can predict trading volume and volatility.

Later, Bollen et al. [9] showed that Twitter mood can even predict stock market trends.

Many recent studies leverage advanced NLP models such as FinBERT, VADER, etc. and connect extracted financial sentiment with returns, volatility, and risk [10–12]. These indeed confirm that textual sentiment can contribute to explaining short-term price movements.

4. Sentiment and Correlation Together

Although there is plenty of work on correlations and plenty on sentiment, very few studies combine both. Several recent papers illustrate that when investor sentiment becomes very negative, correlations between stocks increase, especially during crisis periods [11][12].

However, no generally accepted or standard way of generating sentiment-adjusted correlation matrices exists yet, and the task remains open for future research.

2.2 Research Gaps Identified

From reviewing previous work, several clear gaps appear:

1. Most studies look at the correlations and sentiment separately, not in an integrated system.
2. No single dashboard or tool is currently available that can combine RMT-denoised correlations with sentiment analysis in real time.
3. Despite its behavior deviating from western markets, few studies focus on the Indian stock market.
4. Methods to visualize how sentiment affects cross-stock correlations are still experimental and not standardized.

2.3 How this Gaps are Addressed

This project draws together three significant themes from the literature:

- RMT-based correlation denoising
- Volatility and return analysis
- Sentiment analysis from financial text

By incorporating them into a single system, the project addresses the deficiency of a combined, interpretable pipeline. This also adds practical value since most of these methods are applied to

Indian stocks for which, until recently, research has still been limited. The project visualizes raw versus denoised correlations, sentiment heatmaps, and volatility charts that allow users to clearly see how market structure and sentiment interact-something that previous work hasn't provided in a uniform manner.

CHAPTER 3 : PROBLEM STATEMENT

3.1 Problem Statement

Financial markets generate a huge amount of price data, news, and sentiment every day. Investors often have difficulties understanding how stocks move together, how volatility changes over time, and how market sentiment influences these movements. Traditional correlation matrices are noisy and unreliable, especially in the case of many stocks. At the same time, sentiment analysis is usually studied in separation and seldom combined with structural market analysis.

This will leave a gap in understanding the actual market structure and how investor sentiments impact stock relationships, mainly in the Indian context of financial markets, where research and tools are limited.

Therefore, the problem which this project addresses is:

The objective will be to develop an integrated system that can clean and analyze stock correlations using Random Matrix Theory, measure volatility behavior, extract market sentiment from financial text, and visualize these insights through an interactive dashboard for Indian equities.

3.2 Objectives

1. Analytics & Machine Learning Objectives

- Use **GARCH(1,1)** to calculate past volatility and predict next-day volatility.
- Use **Random Matrix Theory (RMT)** to clean noisy correlation matrices and identify real market patterns.
- Use **FinBERT** to calculate sentiment scores from news and speed it up through caching.
- Compare correlations during high-sentiment and low-sentiment periods.

2. Frontend Objectives

- Add clean and simple visualizations:
    - Plotly charts for volatility
    - Heatmaps for correlations
    - Cards showing insights (sentiment, stress, volatility)
    - RMT explanation panel
- Arrange everything neatly in the Dashboard with an easy-to-understand layout.
- Provide simple explanations so that beginners understand what each chart means.

3. Performance & Reliability Objectives

- Reduce FinBERT loading time using a warmup step.
- Make sure GARCH results are returned in clear, consistent units (percent).
- Ensure all backend data is structured well for the frontend to use easily.
- Store ML models and metrics in the backend for stable performance.

4. Beginner-Friendly UX Objectives

- Show small “insights” that summarise what the numbers mean.
- Add tooltips and plain-language text for things like:
    - volatility
    - market stress
    - correlation strength
    - sentiment
- Make every panel easy to read and understand for first-time users.

CHAPTER 4: METHODOLOGY

This section explains how the project works, the techniques used, the data processing steps, and the overall system design. The goal is to clearly describe the backend, frontend, models, algorithms, and workflow of the system.

4.1 Data Collection

The system works with two types of data:

4.1.1 Stock Price Data

- Collected from **Yahoo Finance** using the Python library **yfinance**.
- Data includes **Open, High, Low, Close, and Volume** for each stock.
- The user selects a stock symbol, date range, and analysis type from the dashboard.

4.1.2 News / Text Data

- News headlines and short articles are fetched from publicly available sources.
- These are used to calculate **sentiment**, which shows whether the news is positive, negative, or neutral.

4.2. Data Preprocessing

Before doing any analysis, the data is cleaned and prepared.

4.2.1 Cleaning Price Data

- Missing values are filled or removed.
- Dates are aligned properly.

- **Daily log returns** are calculated using:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

4.2.3 Cleaning Text Data

- Remove unwanted characters (links, HTML, punctuation).
- Convert to lowercase.
- Split sentences into tokens.
- Remove stopwords.
- Lemmatize words (convert to base form).

The cleaned text is then sent to the **FinBERT sentiment model**.

4.3. Algorithms, Models, and Techniques Used

The project uses four main analytical techniques:

4.3.1 GARCH (Volatility Forecasting)

GARCH(1,1) is used to measure and predict **market volatility**.

Steps:

1. Take daily log returns.
2. Fit a **GARCH(1,1)** model from the arch library.
3. Get:
    - Historical volatility

- Next-day predicted volatility
- Forecast variance
- Model summary

4. Convert results into **percent units** (e.g., 1.5% daily volatility).

This helps users see how risky or stable a stock may be.

4.3.2 Correlation Analysis + Random Matrix Theory (RMT)

To understand how stocks move together, we first compute the covariance.

$$\mathrm{Cov}(X,Y) = \frac{1}{T-1}\sum_{t=1}^{T}(x_t - \bar{x})(y_t - \bar{y})$$

Then the correlation between two stocks:

$$\rho_{ij} = \frac{\mathrm{Cov}(i,j)}{\sigma_i \sigma_j}$$

But correlations often contain a lot of **noise**, so we apply **Random Matrix Theory (RMT)**.

RMT Steps:

1. Compute correlation matrix.
2. Find eigenvalues.
3. Compare with **Marchenko–Pastur limits** (noise bounds).
4. Keep only the meaningful eigenvalues.
5. Build a **denoised correlation matrix**.
6. Show heatmaps for both raw and denoised matrices.

This helps identify true market patterns and stress levels.

4.3.3 Sentiment Analysis (FinBERT)

The project uses the model **FinBERT**, which is trained on finance-related text.

Steps:

1. Send each cleaned news headline to FinBERT.
2. FinBERT returns probabilities for:
    - POSITIVE
    - NEGATIVE
    - NEUTRAL
3. Convert these into a score:
    - Positive → +score
    - Negative → –score
    - Neutral → 0
4. Average all scores for each stock.

Caching

- FinBERT is slow to load, so:
    - The backend supports **/api/warmup-sentiment** to preload it.
    - A small in-memory cache avoids re-scoring the same text.

4.3.4 Sentiment-Adjusted Correlations

This technique checks whether market sentiment changes how stocks move together.

Steps:

1. Split returns by high sentiment vs low sentiment days.
2. Calculate correlation matrices for each case.
3. Compare cluster patterns.

$$C_{ij}^{\text{adj}} = C_{ij} \times \left(1 + \alpha \frac{S_i + S_j}{2}\right)$$

This gives insight into market behaviour during good or bad news periods.

4.4. Tools, Libraries, and Platforms

Backend (Python + Flask)

- Flask (API framework)
- yfinance (price data)
- pandas, numpy (data handling)
- arch (GARCH)
- scipy, numpy.linalg (RMT, eigenvalues)
- transformers (FinBERT)

Frontend (React)

- React.js (UI framework)
- react-plotly.js + plotly.js (charts)
- Tailwind-style classes (styling)

- Custom React components:
    - GarchPanel
    - InsightsCard
    - RmtInterpretation
    - Heatmap containers

Environment / Dev Tools

- VS Code
- Git & GitHub

4.5. System Workflow (Simple Description)

Backend Flow

1. Receive a request from frontend.
2. Fetch price data & news.
3. Preprocess the data.
4. Run analytics:
    - RMT
    - GARCH
    - Sentiment
5. Package everything into JSON.
6. Send results to frontend.

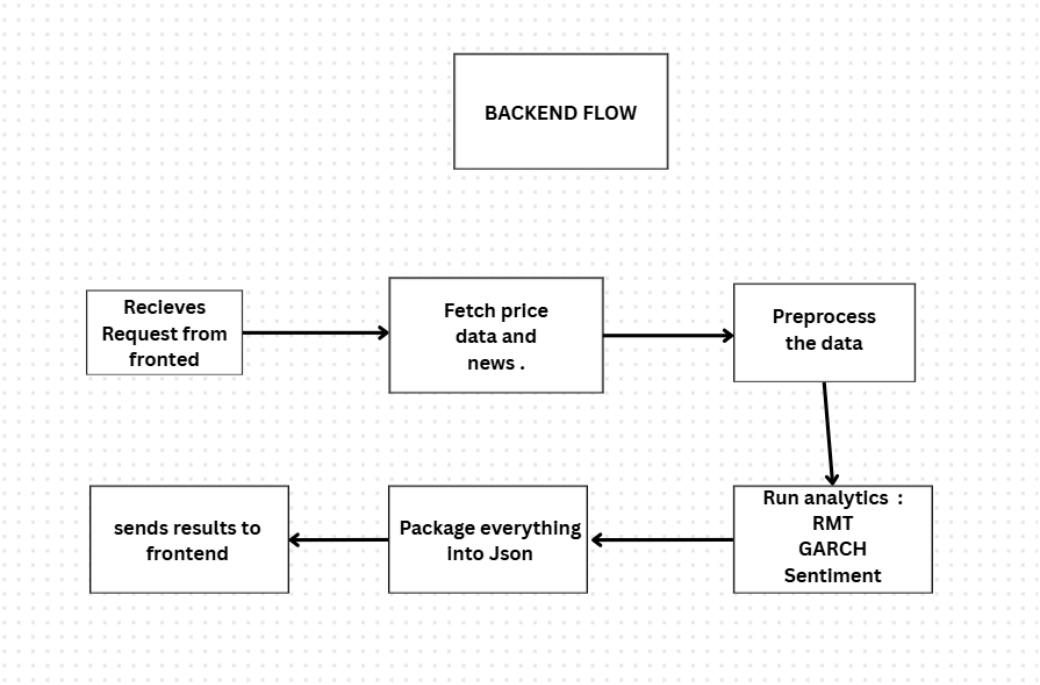

Fig 4.5.1: Backend Flow

Frontend Flow

1. The user selects a stock or opens the dashboard.
2. Dashboard sends requests to `/api/analyze` or `/api/garch_volatility`.
3. Plotly charts render results:
    - Volatility chart
    - Correlation heatmap
    - Sentiment score
    - Insights card
    - RMT interpretation
4. UI is shown in clean, simple language.

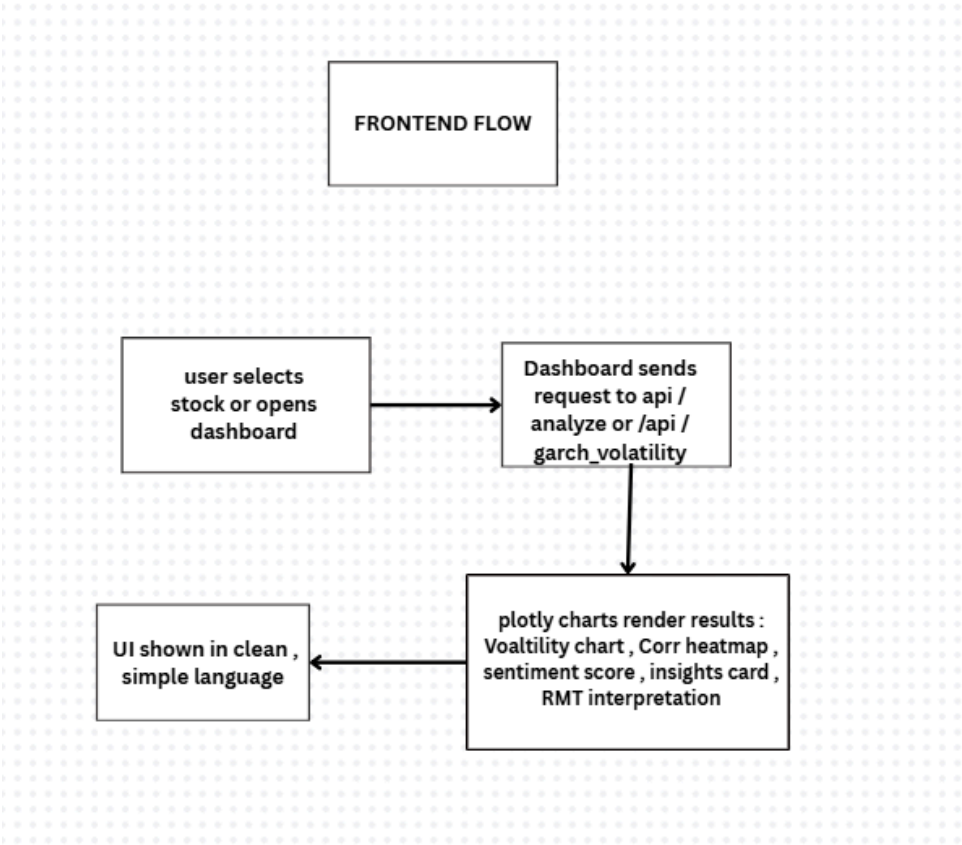

Fig 4.5.2: Frontend Flow

4.6. System Architecture

Backend (Flask)

- Handles API requests
- Runs ML models (GARCH, FinBERT)
- Runs analytics (RMT, correlations)
- Caches data and sentiment scores
- Sends response to frontend

Frontend (React)

- Displays charts and results
- Calls APIs
- Shows insights and interpretations
- Provides a beginner-friendly interface

Together, the system forms a complete “analysis pipeline”.

CHAPTER 5 :EXPERIMENTAL/SIMULATION/ANALYTICAL ANALYSIS AND DISCUSSION OF RESULTS / DASHBOARD

This chapter is divided into two main parts:

1. Dashboard-Based Results (from INDstocks clone)

2. Research-Based Analytical Results (from our backend + ML work)

5.1 Dashboard-Based Results

This part shows the results and UI components that were obtained from the INDstocks frontend clone.
These pages show stock charts and overview information, and serve as the foundation for the analytics extension.

5.1.1 Main Dashboard Page

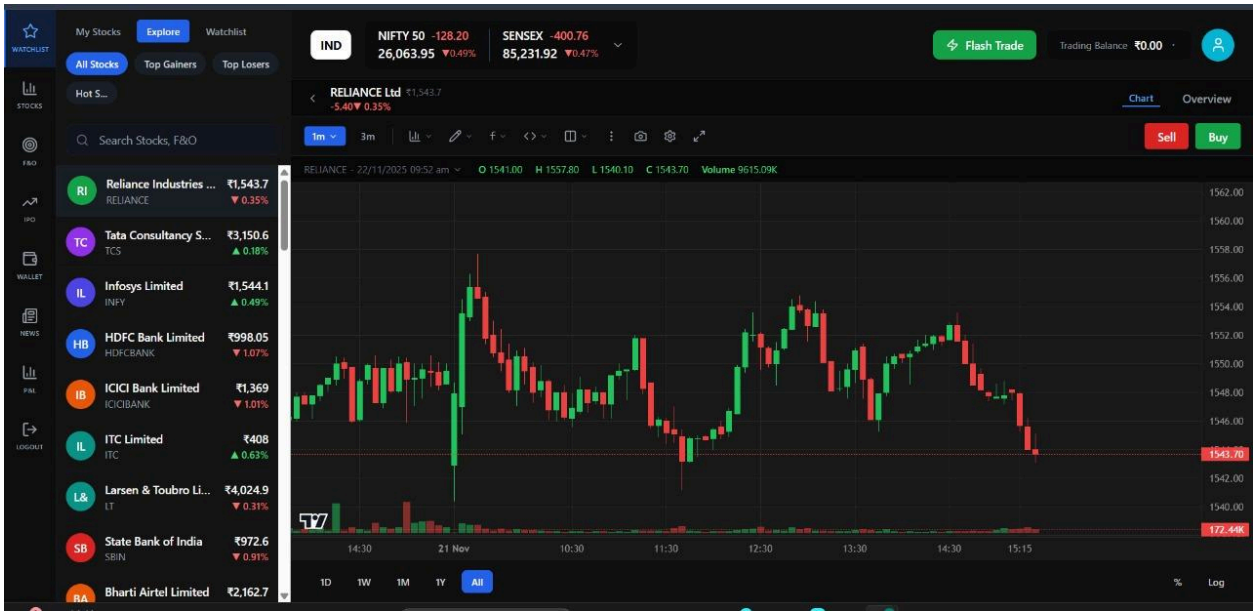

Fig 5.1.1 Main Dashboard Page

Candlestick chart interface showing real-time price movement of *Reliance Industries Ltd* along with market overview, stock list, and trading options as part of the cloned INDstocks dashboard.

The screen features several interactive components:

1. Market Indices (Top Bar)
    - The real-time values for NIFTY 50 and SENSEX, including the percentage change, are shown.
    - This provides an indication of market direction (both values are negative in this image).
2. Stock Watch-list (Left Sidebar)
    - The stock watch-list displays those stocks likely to be monitored - for example, Reliance, TCS, Infosys, HDFC Bank, ICICI Bank.
    - Each stock is given, the current market price and dollar and percentage change.
    - This will facilitate switching to other stocks quickly.
3. Selected Stock Panel (Top Block) : In this case, the selected stock is Reliance Industries Ltd. The screen shows:
    - Current price (₹ 1,543.7)
    - Daily change (−5.40, −0.35%)
    - Additional options to select the time span (1m, 5m, 1h, etc.)
4. Main Candlestick Chart (Middle Block)
    - Shows price variation over short time-frame intervals.
    - Green candles show an increase price over the previous interval while red candles show a weaker price over the previous interval.
    - Vertical bars show intraday (i.e. in the trading day) price volatility and volume.
    - Timestamp axis shows time of intraday trading (9:52 to 15:20).
5. Volume Bars (Bottom of Chart)
    - Show the number of shares traded (volumes traded) in the interval price activity.
    - Higher bars traditonally show where there were faster prices around those bars.

5.1.2 Top Gainers And Losers

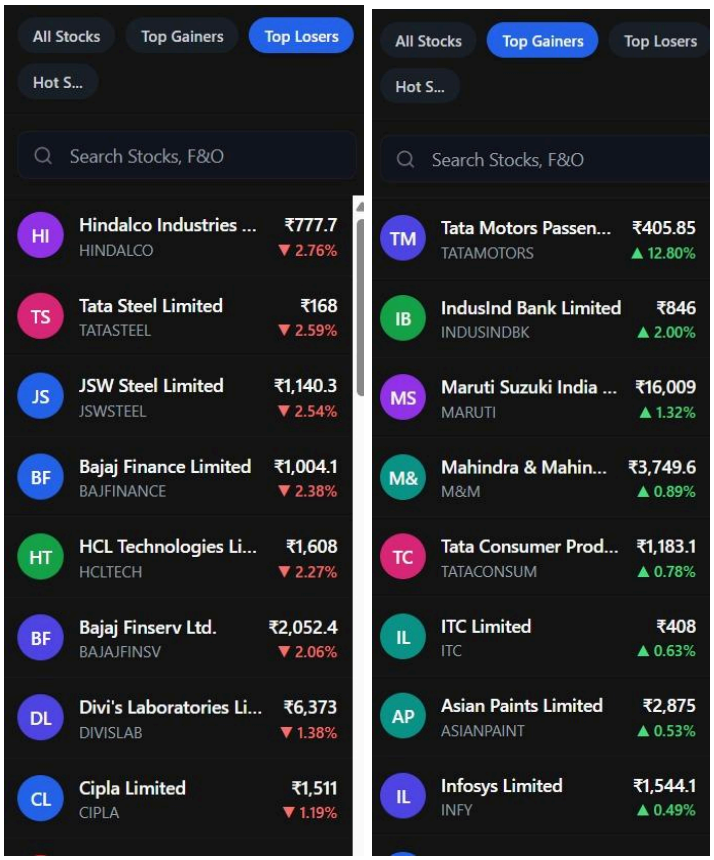

Fig 5.1.2 Top Losers and Gainers

Two screenshots of the Top Gainers and Top Losers lists are shown here from the cloned INDstocks dashboard. These panels serve as a quick summary of market action for the user.

- The Top Gainers list includes stocks with the largest percentage gain (or increase) for the trading session. Example: Tata Motors (+12.80%), IndusInd Bank (+2.00%), Maruti (+1.32%).
- The Top Losers list displays stocks with the largest percentage 'loss' (or decline). Example: Hindalco (−2.76%), Tata Steel (−2.59%), JSW Steel (−2.54%).

Each entry includes:

- Company name and stock ticker
- Current Market price
- Percentage change (green arrow for increase, red arrow for a decline)
- A colored icon for easy identification (visual identification)

This is all part of the INDstocks UI, and it is a perfectly cognised "snapshot" of market action in regards to behaviour of stocks in the overall market.

5.1.3 Overview Page

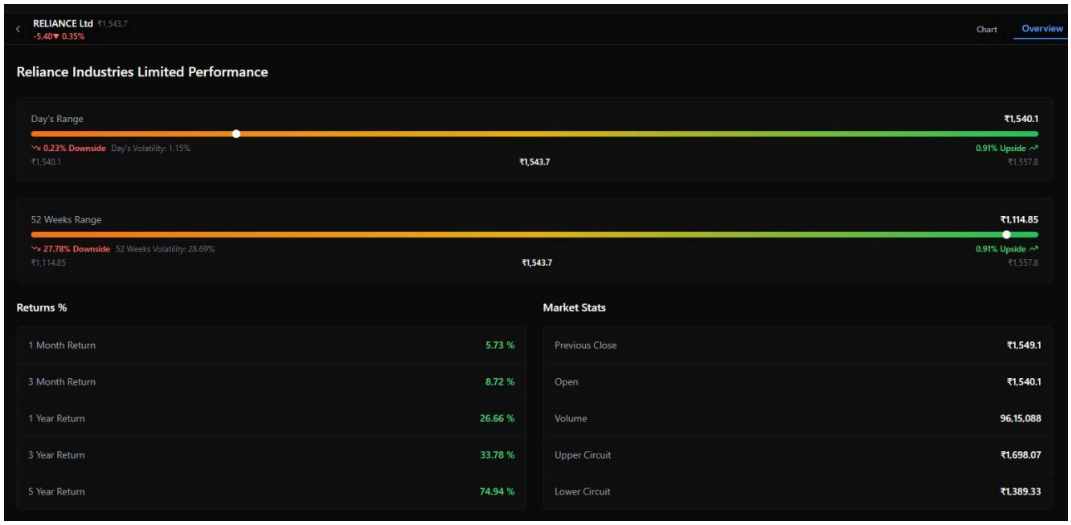

Fig 5.1.3 Overview Page

This picture shows the Overview tab for Reliance Industries Ltd through the replicated INDstocks interface. This page is designed to provide an easy summary of the stock's performance based on timeframes, and market conditions.

The page consists of a few key sections.

1. Day's Range Indicator

This shows the lowest and highest price for the stock's performance on the current trading day.

The colour bar ranges between red (downside) and green (upside).

Indicates:

- Current day's volatility (~1.15%)
- Current market price (₹1,543.7)

This gives users some idea of intraday price fluctuations.

2. 52-Week Range Indicator

- This shows the lowest and highest price for the stock's performance in the past year.
- Again, the bar ranges between red (close to the low) and green (close to the high).

- Indicates 52-week volatility (~28.69%)
- Users can use this information to assess long-term performance, and if the stock is closer to its yearly high or low.

3. Returns (%) Section

The left panel shows historic returns as a percentage:

- 1 Month Return: 5.73%
- 3 Month Return: 8.72%
- 1 Year Return: 26.66%
- 3 Year Return: 33.78%
- 5 Year Return: 74.94%

This panel gives users ease of reading, of periodic performance.

4. Market Stats Section

The right panel demonstrates market and trading fundamentals like:

- Previous Close
- Opening Price
- Volume
- Upper Circuit
- Lower Circuit

These performance fundamentals can help traders and investors.

5.2 Research Based Results

5.2.1 Correlation Matrices

Correlation (raw)

	HDFCBANK.NS	ICICIBANK.NS	INFY.NS	TCS.NS
HDFCBANK.NS	1.00	0.46	0.23	0.19
ICICIBANK.NS	0.46	1.00	0.25	0.20
INFY.NS	0.23	0.25	1.00	0.72
TCS.NS	0.19	0.20	0.72	1.00

Correlation (sentiment-adjusted)

	HDFCBANK.NS	ICICIBANK.NS	INFY.NS	TCS.NS
HDFCBANK.NS	1.00	0.49	0.26	0.21
ICICIBANK.NS	0.49	0.97	0.26	0.21
INFY.NS	0.26	0.26	1.00	0.79
TCS.NS	0.21	0.21	0.79	1.00

Fig 5.2.1 Raw and Sentiment Adjusted Correlation Matrices

Raw Correlation Matrix

This figure represents the raw correlation matrix generated from the project's backend. Each cell shows how closely the returns of two stocks move together:

- Values closer to 1 indicate a very strong positive correlation.
- Values near 0 indicate weak or no correlation.
- Colours represent correlation strength (light green = low, bright green = high).

In this raw matrix:

- HDFC Bank and ICICI Bank show moderately high correlation (0.46).
- INFY and TCS show a very high correlation (0.72), which is expected because both are IT sector stocks.
- Cross-sector pairs (Bank vs IT) show lower correlations (0.19–0.25 range).

Interpretation:

This raw matrix shows natural sector grouping:

- Banks correlate strongly with other banks

- IT companies correlate strongly with other IT companies
- Cross-sector correlations are lower

This confirms sector-based movement in the Indian market. However, the raw correlation may contain noise due to short data windows, market noise, and random fluctuations.
Hence, the project also computes a sentiment-adjusted correlation matrix.

Sentiment Adjusted Matrix

This figure shows the correlation matrix after adjusting for sentiment, meaning the correlations were calculated using only the days with similar sentiment classification (e.g., low/negative sentiment).

Key observations:

- HDFC Bank and ICICI Bank correlation slightly increased (from 0.46 → 0.49).
- INFY and TCS correlation increased (0.72 → 0.79).
- Cross-sector correlations (Banks ↔ IT) also increased slightly (e.g., 0.23 → 0.26).

This demonstrates a known market behaviour:
When sentiment becomes negative or uncertain, stocks tend to move together more strongly.

Interpretation:

The sentiment-adjusted matrix clearly shows:

- Correlation increases during negative sentiment periods
- Fear or uncertainty causes investors to react similarly across stocks, even across sectors
- Market becomes more “clustered” or “tightly coupled” during bad-news conditions
- IT sector tightens more significantly (0.79 correlation between INFY and TCS)

This supports the concept that market sentiment plays a major role in price co-movement, and correlational patterns change depending on the emotional tone of the market.

5.2.2 Quick Insights

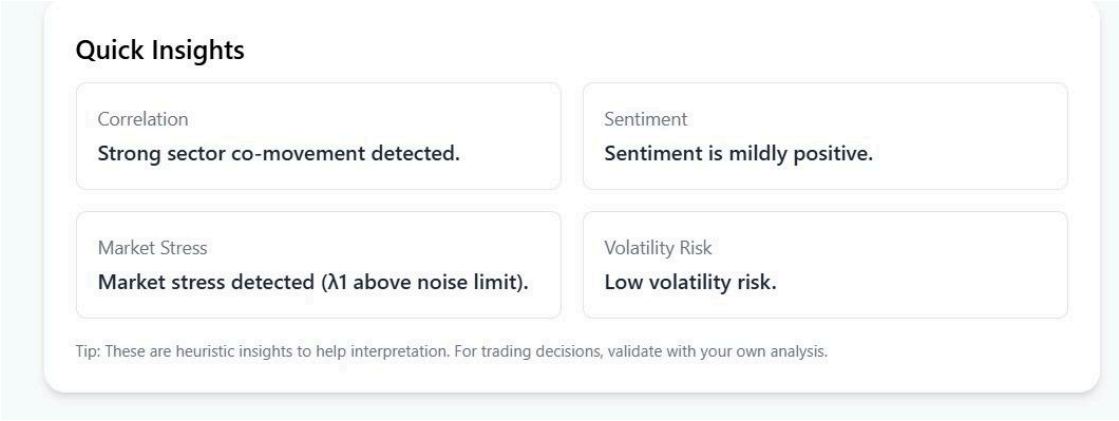

Fig 5.2.2 Quick Insights Panel

The insights engine analyzes numerical results and produces simple interpretations:

- Correlation: Strong sector co-movement
- Sentiment: Mildly positive
- Market Stress: λ_1 above noise limit = stress detected
- Volatility Risk: Low risk

These insights help beginners interpret complex analytics.

5.2.3 Some Visualizations

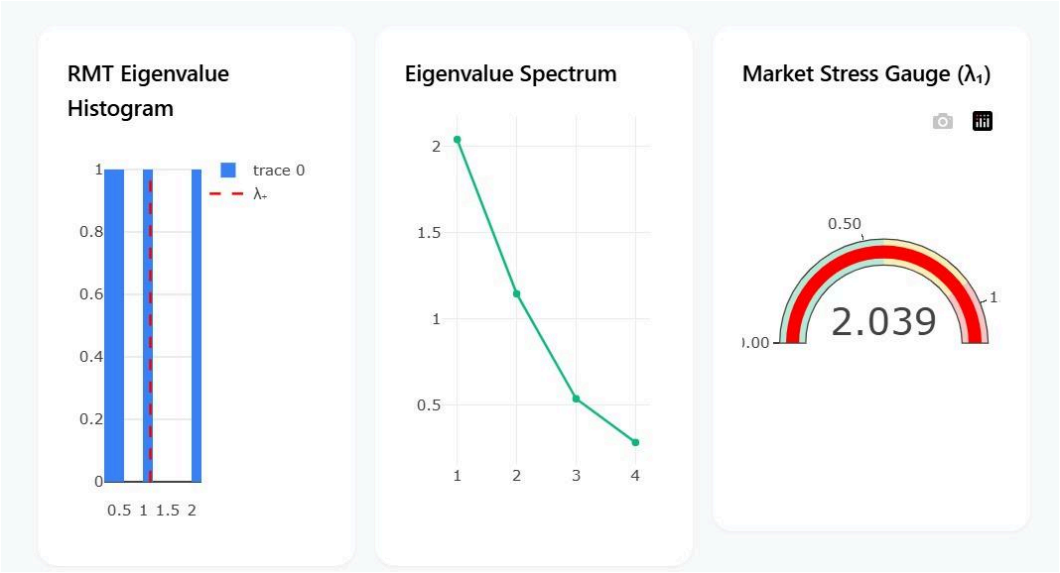

Fig 5.2.3 Eigen Value Spectrum and Stress Gauge

EigenValue Histogram:

This histogram displays eigenvalues of the correlation matrix. The red dashed line marks the Marchenko–Pastur upper bound (λ^+).

If eigenvalues exceed λ^+, they represent true market structure (e.g., sector clustering or market-wide movement). If they fall within the noise band, they are considered random noise.

EigenValue Spectrum:

The eigenvalue spectrum shows how quickly eigenvalues decline.

A large first eigenvalue followed by smaller ones means that **most variance** is explained by one dominant factor (market effect).

Stress Gauge:

Gauge chart showing market stress level based on the ratio of the largest eigenvalue (λ_1) to the RMT noise limit (λ^+).

If $\boldsymbol{\lambda_1 > \lambda^+}$, market stress is present (stocks move together more than usual).
This gauge provides a simple visual display of stress intensity.

In this screenshot, the gauge reads 2.039, which is above the noise threshold.

A value above 1 indicates stress or strong co-movement.

Here, λ_1 exceeds the noise limit, meaning the market is experiencing moderate to high stress, causing stocks to behave similarly.

5.2.4 Hybrid ARIMA GARCH Forecast

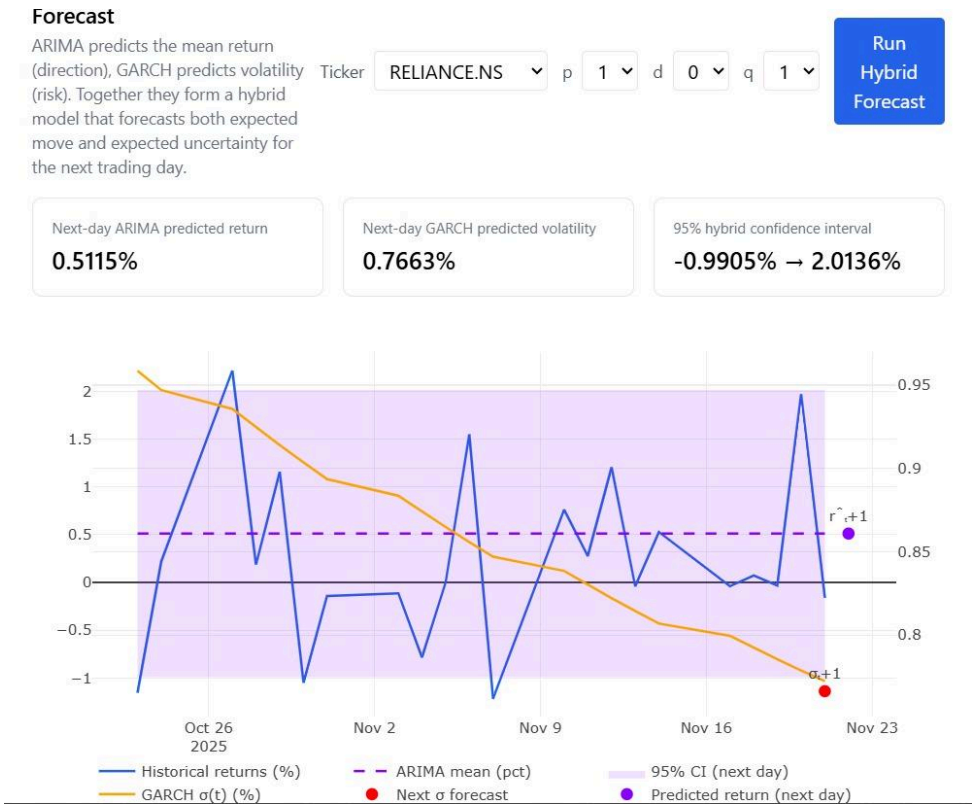

Fig 5.2.4 Hybrid ARIMA-GARCH Forecast

This figure displays the hybrid forecasting component of the item. The hybrid forecasting component utilizes:

- ARIMA to predict the next-day mean return of

- GARCH to predict the next-day volatility risk of (σ_{t+1})

- A 95% combined confidence interval to illustrate expected uncertainty both in benign terms and to facilitate expected volatility

The chart includes:

- Blue line → Historical daily returns

- Yellow line → GARCH conditional risk volatility

- Purple dot → ARIMA predicted return

- Red dot → Day-forward risk volatility prediction

- Shaded region → 95% Hybrid CI

- Dashed purple line → ARIMA mean

In this example ARIMA predicts a next-day return of +0.5115% while GARCH predicts a next-day profit claim risk (volatility) of 0.7663%.

In other words:

The visual aids the user reconstruct their understanding of both financial return expectations and potential risk direction. The forecast depicts a moderate yet positive expected return direction with low-to-medium implied uncertainty. The relatively narrow confidence interval shows that volatility has been relatively stable recently.

CHAPTER 6: CONCLUSIONS

The project illustrated that advanced financial concepts such as volatility modelling, sentiment extraction, Random Matrix Theory, and correlation structure can be combined in one analytics system that is easy to use for someone who is new to finance. By combining interactive GARCH volatility forecasts, FinBERT sentiment scores, RMT eigenvalue structure and correlation visualisation, we were able to demonstrate that sometimes, complex market behaviour can be understood better than "over-thinking" it.

The analytics system provided meaningful insights about market behaviour such as periods of stress, market co-movement, volatility clustering and the effect of sentiment changes on correlations. As impressive as the system is, we still faced limitations with data reliabilities, computational costs with GARCH and FinBERT sentiment analysis, and the scalability of this project. Perhaps with the next iteration, network analysis and or advanced ML models could be incorporated to move toward a more robust market intelligence assistant.

6.1 Contributions of the Project

1. **Utilized advanced analytics on a single platform**: The project incorporates GARCH volatility, sentiment analysis, correlation matrices, and Random Matrix Theory—models that we typically treat independently—into a unified dashboard.
2. **Simplified multi-layered financial constructs**: Technical constructs such as eigenvalue spectrum, λ^+ thresholds, volatility clustering and sentiment-prompted correlation shifts are all represented simply and visually, with the aim of simplifying the concepts for beginners.
3. **Developing Automated interpretations**: The system takes numerical outputs and interprets them into plain-language, allowing users to deeply and accurately understand market stress, sentiment polarity, and volatility signals without prior domain experience.
4. **Streamlined backend efficiency**: Price caching, warming sentiment models and fast endpoint APIs will minimize latency while allowing for a better user experience.
5. **Developed clear and interactive UI elements:** The dashboard provides heatmaps, vol curves, sentiment indicators, eigenvalue charts, and predictions, all in an organized and clear layout.

6.2 Major Findings

- **Markets exhibit clustered volatilities:** GARCH (1,1) results demonstrate that volatility is not randomly generated but instead, experiences periods of low followed by sustained increases in volatility.

- **RMT effectively reduces noise**: The eigenvalue spectrum indicates that a few of the larger eigenvalues convey information, and the majority are simply random noise. The highest eigenvalue (λ_1) serves as a consistent leading indicator of turbulent conditions for markets.
- **Sentiment influences correlation behavior:** Sentiment-adjusted correlations indicate that in periods of extremely positive or negative sentiment, stock co-movement raises and falls at exaggerated rates compared to the price data alone.
- **Combined analytics provide a deeper understanding**: Employing sentiment, volatility, and RMT collectively enables users to better adjust and identify market regimes. For an example:
    - High λ_1 + negative sentiment → early potential stress
    - High sentiment + increasing correlations → aggressive bullish herding
    - Low correlations + low volatility → stable regime
- **Easy for beginners to interpret the markets**: Using visual summaries and automated voices permits to interpret advanced concepts without having to repeatedly compute mathematics.

6.3 Limitations

- **Reliance on external sentiment data:** Sentiment analysis relies on data sourced from external news sources or social media, which may consist of biased, partial, or delayed information.
- **GARCH model computational cost**: Fitting a GARCH model to a larger dataset or across multiple tickers can incur significant costs and slow down real-time applications.
- **Random Matrix Theory has a data requirement for stocks**: If the number of stocks (or historical window) is small, the estimates of the eigenvalues can be unreliable.
- **FinBERT takes time to run**: While FinBERT is cached, it still takes time to run the first model and performs worse without GPU.
- **Not a predictive trading system:** The system provides insights and analysis, not guaranteed buy/sell signals or financial advice.

6.4 Future Recommendations

1. **Include network analysis**: Utilizing Minimum Spanning Tree (MST), centrality measures, and community detection may uncover latent structures and sector correlations.
2. **Utilize more advanced forecasting models**: Models such as LSTM, Transformer-based volatility forecasting, or GARCH-MIDAS provide a more sophisticated method for predicting volatility than the standard forecast models for long-term volatility.

3. **Implement real-time streaming data**: Combine websockets or real-time APIs to display live volatility, sentiment movements, and correlation shifts over time while the market moves.
4. **Enhance sentiment modelling**: Incorporate multiple sentiment sources (news, X/Twitter, Reddit) and weigh based on quality to improve sentiment scoring.
5. **Host the system on cloud infrastructure**: This improves scalability and allows several users to have access to the system at the same time.
6. **Include explainability elements**: To provide users who are inexperienced with the system context regarding which predictions and stress indicators appear, include a simple language explanation and an AI-generated summary.

CHAPTER 7: CONTRIBUTION OF MEMBERS

Ananya-

- Worked on the frontend using React.
- Built charts and visual components such as heatmaps, volatility graphs, eigenvalue plots, and the insights panel.
- Styled the dashboard and improved the UI.
- Added Methodology and Results to the report.

Preet-

- Worked on the backend using Flask.
- Did data preprocessing for price and news data.
- Added main analysis features like GARCH volatility, RMT correlation, and sentiment analysis.
- Connected all backend APIs with the frontend dashboard.
- Helped write the technical sections of the report.

Mayank-

- Work on fetching the the data of all stocks in graphs
- Helped with documentation, formatting, and final presentation.
- Worked on the backend using Flask and Frontend for cloning INDStocks (Overview, Candlestick chart, Gainers/Losers, Watchlist, graph).
- Work on data fetching from yfinance.

Himanshi-

- Did data preprocessing for price and news data and added RMT correlation and sentiment analysis.
- Helped in creating sentiment-adjusted correlations.
- Contributed to the research-based results in the report.
- Integrated the INDstocks cloned dashboard pages (Overview, Candlestick chart, Gainers/Losers, Watchlist).
- Helped with documentation, formatting, and final presentation.

CHAPTER 8: REFERENCES

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